

b)	Code sequence: <table border="1"> <thead> <tr> <th>Instruction</th> <th>Machine Code of the corresponding instruction converted into Hex.</th> </tr> </thead> <tbody> <tr> <td>Label0: SUB X10, X9, X23</td> <td></td> </tr> <tr> <td>Label1: BNE X24, X25, Label3</td> <td></td> </tr> <tr> <td>Label2: ADD X5, X5, X6</td> <td></td> </tr> <tr> <td>SUB X5, X22, X5</td> <td></td> </tr> <tr> <td>Label3: SLLI X22, X22, 3</td> <td></td> </tr> <tr> <td>ADD X22, X22, X23</td> <td></td> </tr> <tr> <td>Label4: LD X6, 32(X22)</td> <td></td> </tr> <tr> <td>BEQ X4, X8, _____</td> <td>0xFE825A80</td> </tr> <tr> <td>Label5: SW X7, 8(X21)</td> <td></td> </tr> </tbody> </table> Find the correct label to be placed in the blank. Answer this in your script. You must show the necessary calculations in your script.	Instruction	Machine Code of the corresponding instruction converted into Hex.	Label0: SUB X10, X9, X23		Label1: BNE X24, X25, Label3		Label2: ADD X5, X5, X6		SUB X5, X22, X5		Label3: SLLI X22, X22, 3		ADD X22, X22, X23		Label4: LD X6, 32(X22)		BEQ X4, X8, _____	0xFE825A80	Label5: SW X7, 8(X21)		3
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4. CO2	Translate the following codes written in high level language into an instruction sequence written in RISC-V Assembly. Assume that variables i, j, and n are stored in registers X21, X22, and X23. A and B are <u>byte arrays</u>. A[0] and B[0] are stored in X5 and X6 respectively.																					
a)	<pre> if (A[i] <= 3) { if (B[j] != 4) { B[j] -= A[i] } else { B[j] += 5 } } </pre>	3																				
b)	<pre> i = -3; n = 7; while (n >= i) { n -= 1; } </pre>	2																				
5. CO1	a) You are asked to store the binary number -1.1×2^{-30} in an IEEE format. What is the minimum number of bits you need to reserve for the exponent field?	2																				
b)	You are given the following IEEE representation where <u>7 bits</u> are used to store the exponent. Convert 0xABCD to a binary normalized floating point number.	3																				

1.
CO1

A processor executes a total of 30×10^9 instructions divided among the following types of instructions. The distribution of instruction types and their corresponding clock cycles are as follows:

Instruction Type	Instruction Count	Cycle per instruction
R-type	20% of total instructions	2
I-type	25% of total instructions	1
SB-type	30% of total instructions	2
U-type	25% of total instructions	1

a) Find the average CPI of the instructions.

2

b) Calculate the execution time of the given instructions, given the CPU has a frequency of 3.5 GHz.

3

2. a) Read each statement carefully. Determine if the following statements are **True or False**. If you select False, you must provide the correct version of the statement.

i) and X21, X22, X23

The instruction can be expressed using the following format:

2 type

xxxx xxx	rs1	rs2	xxx	rd	xxxx xxx
7	5	5	8 3	5	7

ii) Is the following RISC-V instruction correct for implementing $j = A[5]$, given that j, and the base address of the half-word array, A, are stored in X21, and X19, respectively?

sd X21, 20(X19)

iii) It is possible to jump from the address 0xA0A0FAFC to 0xA0A0FB02 using SB-type instructions.

b)
CO1

CPU A takes 100 seconds to execute 300 instructions, with each instruction requiring the same amount of execution time. [Now, CPU B can make 60% of the instructions faster by 2 times] Calculate the total time required by CPU B to execute all instructions.

3.
CO2

a) Load the number 0x00DBE2F1 in the register x22.

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